

Dimension dependence in Hermite–Hadamard inequalities

Literature answer to questions in arXiv:1905.03216

fa_banach_001 · lane 7 · 17 August 2026

Disposition. Both qualitative dimension questions are already answered negatively by the direct follow-up arXiv:1907.06122. No new mathematical result is claimed.

1. Exact questions

For convex bodies $\Omega \subset \mathbb{R}^n$, the source first considers the normalized inequality

$$\frac{1}{|\Omega|} \int_{\Omega} f \, dx \leq \frac{A_n}{|\partial\Omega|} \int_{\partial\Omega} f \, d\sigma \quad (1)$$

for admissible convex functions. Immediately after its Theorem 1, it asks whether the optimal A_n remain bounded with dimension. Its scale-invariant theorem concerns positive subharmonic functions and constants B_n in

$$\int_{\Omega} f \, dx \leq B_n |\Omega|^{1/n} \int_{\partial\Omega} f \, d\sigma; \quad (2)$$

the same paragraph asks whether B_n might decay as $n \rightarrow \infty$ [1].

The queue’s literal phrase “we do not know” is not an open problem: it only says that one step of the Brownian-motion proof does not track the positions of particles surviving to a fixed time.

2. Direct literature answer

Literature answer. *The direct follow-up [2] proves that the optimal constants in (1) satisfy*

$$A_n \geq \max\{n-1, 1\}.$$

Consequently no dimension-free constant exists. It also identifies, on a fixed convex domain, the optimal constant in $\int_{\Omega} f \leq c(\Omega) \int_{\partial\Omega} f$ over positive subharmonic f as

$$c(\Omega) = \max_{x \in \partial\Omega} \partial_{\nu} u(x), \quad -\Delta u = 1, \quad u|_{\partial\Omega} = 0,$$

where ν is the inward normal.

The lower-bound construction for A_n uses positive convex functions and therefore lies inside the class asked about in the source. The fixed-domain characterization follows by Green’s identity for the upper bound and by Poisson extensions concentrating at a maximizer for the reverse inequality.

3. Certified non-decay of the scale-invariant constants

The ellipsoid family displayed in the source already gives the qualitative answer to the second question, after one normalization correction. Put

$$E_n = \left\{ x : \frac{x_1^2}{4} + \sum_{j=2}^n \frac{x_j^2}{2n-2} < 1 \right\}, \quad q_n = 1 - \frac{x_1^2}{4} - \sum_{j=2}^n \frac{x_j^2}{2n-2}.$$

Direct differentiation gives $-\Delta q_n = 3/2$, although the source calls q_n the torsion function. Hence the correctly normalized torsion function is $u_n = (2/3)q_n$. At $p = (2, 0, \dots, 0)$,

$$\partial_\nu u_n(p) = \frac{2}{3}.$$

Moreover,

$$|E_n|^{1/n} = \left[\omega_n 2(2n-2)^{(n-1)/2} \right]^{1/n} \longrightarrow 2\sqrt{\pi e}.$$

The fixed-domain characterization therefore yields

$$\liminf_{n \rightarrow \infty} B_n \geq \frac{1}{3\sqrt{\pi e}} > 0.$$

Thus the optimal constants in (2) do not decay to zero. The follow-up explicitly records the same qualitative conclusion and improves the upper bound to $B_n \leq (\omega_n^{1/n} \sqrt{n})^{-1} \rightarrow (2\pi e)^{-1/2}$.

4. Verification and scope

The cited theorem and fixed-domain proposition were checked in the local source of the follow-up. The Laplacian, normal derivative, ellipsoid volume, and Stirling limit above were recomputed independently. The normalization slip only changes the numerical lower constant; it does not alter non-decay. This packet is a literature audit, not a novel proof packet.

References

- [1] J. Lu and S. Steinerberger, *A Dimension-Free Hermite–Hadamard Inequality via Gradient Estimates for the Torsion Function*, arXiv:1905.03216 (2019).
- [2] T. Beck et al., *Improved Bounds for Hermite–Hadamard Inequalities in Higher Dimensions*, arXiv:1907.06122; J. Geom. Anal. 31 (2021), 801–816.